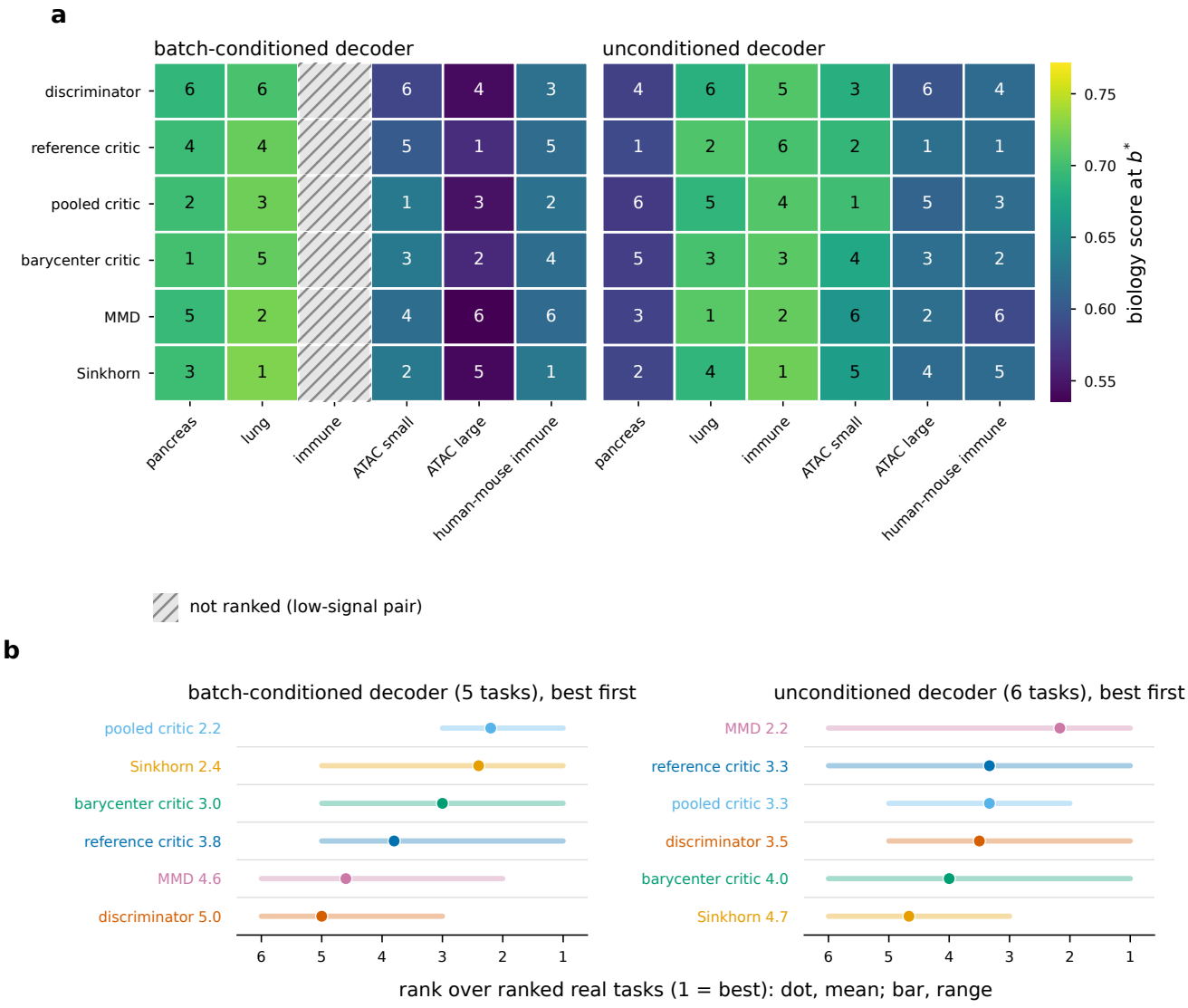

a
batch-conditioned decoder
unconditioned decoder
discriminator
reference critic
pooled critic
barycenter critic
MMD
Sinkhorn
6 6 6 4 3
4 4 5 1 5
2 3 1 3 2
1 5 3 2 4
5 2 4 6 6
3 1 2 5 1
4 6 5 3 6 4
1 2 6 2 1 1
6 5 4 1 5 3
5 3 3 4 3 2
3 1 2 6 2 6
2 4 1 5 4 5
pancreas
lung
immune
ATAC small
ATAC large
human-mouse immune
0.75
0.70
0.65
0.60
0.55
biology score at b*
not ranked (low-signal pair)
b
batch-conditioned decoder (5 tasks), best first
pooled critic 2.2
Sinkhorn 2.4
barycenter critic 3.0
reference critic 3.8
MMD 4.6
discriminator 5.0
6 5 4 3 2 1
unconditioned decoder (6 tasks), best first
MMD 2.2
reference critic 3.3
pooled critic 3.3
discriminator 3.5
barycenter critic 4.0
Sinkhorn 4.7
6 5 4 3 2 1
rank over ranked real tasks (1 = best): dot, mean; bar, range